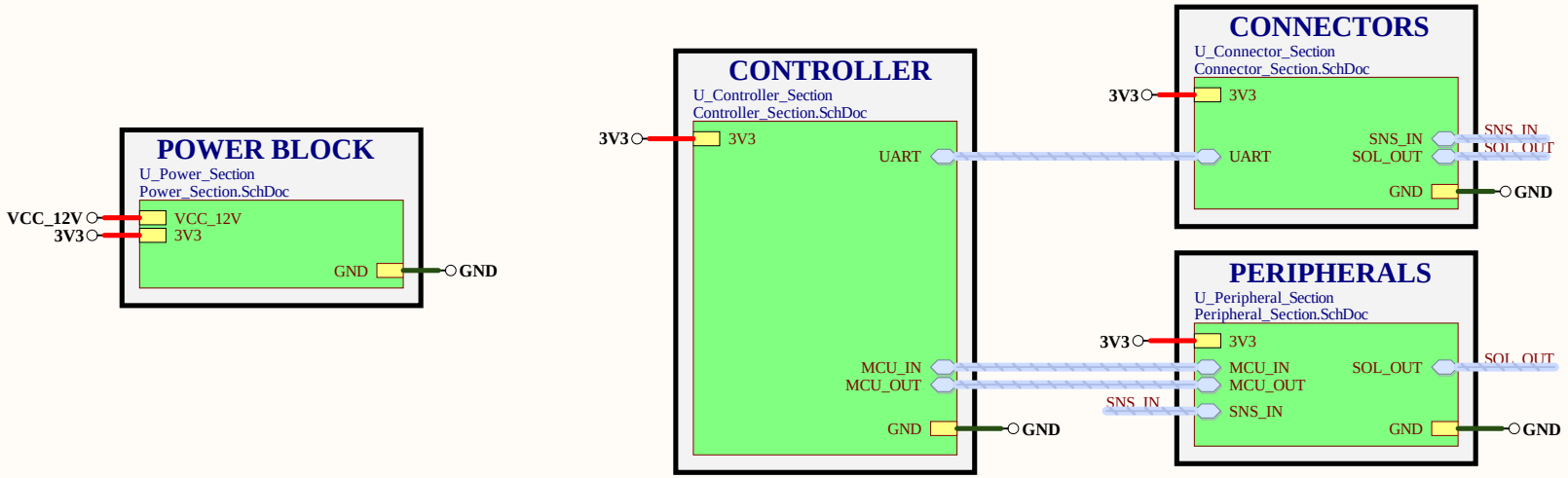
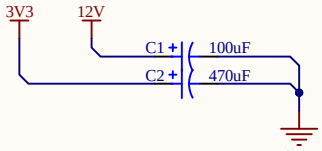
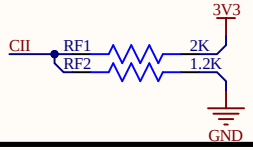

POWER BLOCK
U_Power_Section
Power_Section.SchDoc
VCC_12V
3V3
VCC_12V
3V3
GND
GND
CONTROLLER
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
UART
MCU_IN
MCU_OUT
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
3V3
3V3
UART
SNS_IN
SOL_OUT
SNS_IN
SOL_OUT
GND
GND
PERIPHERALS
U_Peripheral_Section
Peripheral_Section.SchDoc
3V3
3V3
MCU_IN
MCU_OUT
SNS_IN
SOL_OUT
SOL_OUT
GND
GND

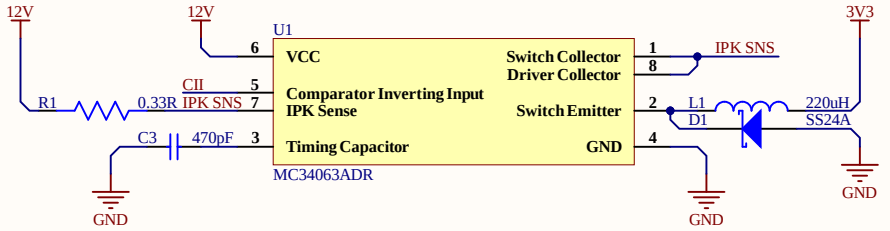
DECOUPLING CAPS



FEEDBACK NETWORKS

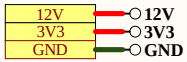


12-3.3V BUCK CONVERTER



VOUT = 1.25 (1+ (R2/R1))=
R2=1.96K (datasheet pg:15)

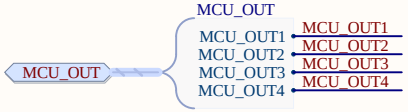
Output current: 500mA



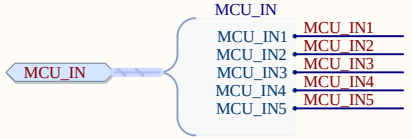
UART SIGNALS



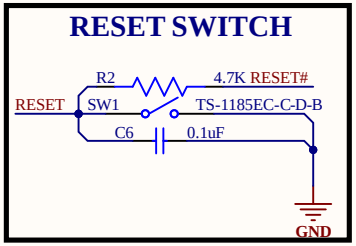
SOLENOID CONTROL SIGNAL



CONTROLLER SENSE INPUT

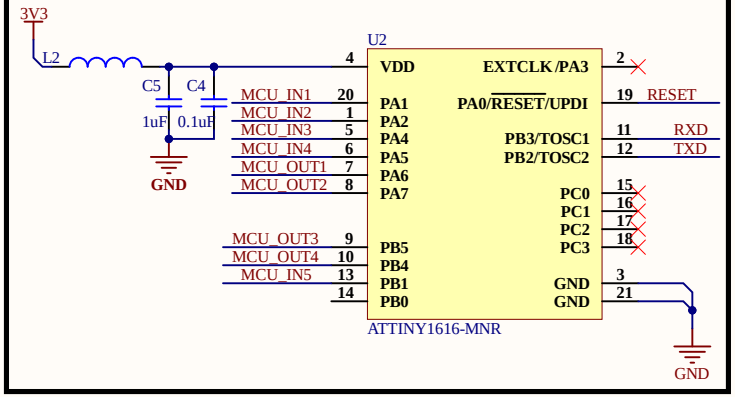


RESET SWITCH

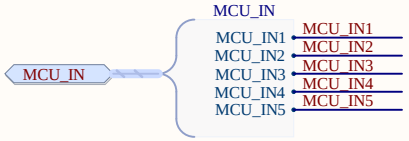




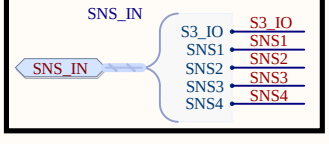
CONTROLLER



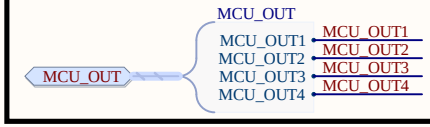
CONTROLLER SENSE INPUT



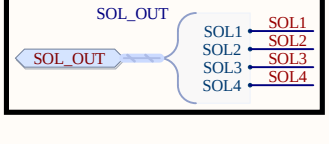
LEVEL SENSE INPUT



SOLENOID CONTROL SIGNAL



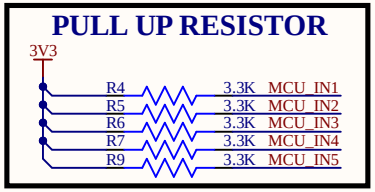
SOLENOID OUTPUT



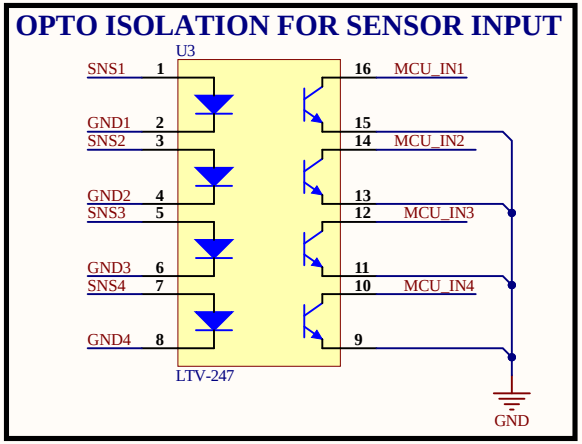
MCU is controlling the base of MOSFETS
VGATE threshold= 2.7 (pg: 2)
Solenoid Current consumprtion=300mA (pg:4)
Solenoid will be operated at 12V

When SNS input=0V,
MCU will sense 3.3V
When SNS=12V, MCU
voltage will sense GND

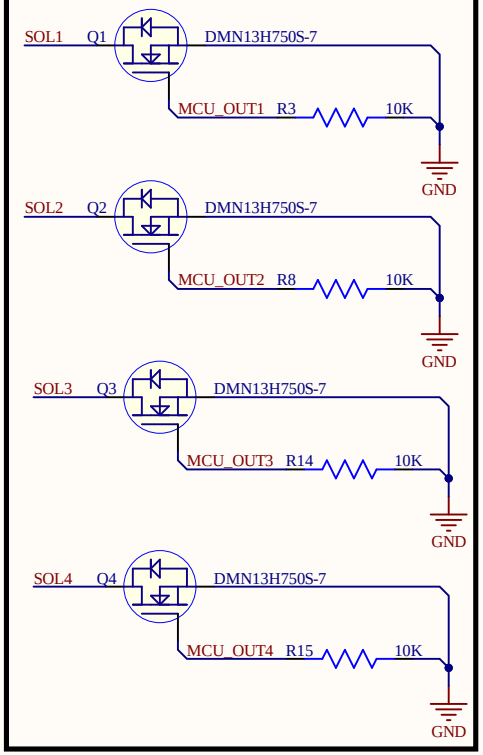
PULL UP RESISTOR



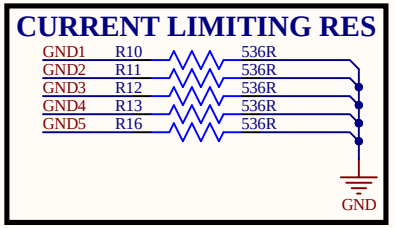
OPTO ISOLATION FOR SENSOR INPUT



SOLENOID CONTROL CIRCUIT



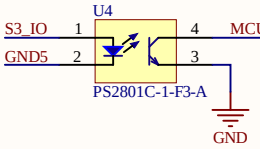
CURRENT LIMITING RES





VF=1.2 (nominal) datasheet pg:8
IF=20mA
VRES= 12-1.2= 10.8V
RES= 10.8V/0.02A=530R
P= (0.02)^2*536R=0.22W

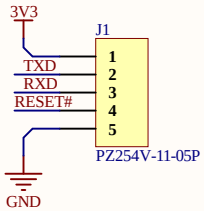
OPTO ISOLATION FOR S3 INTERRUPT



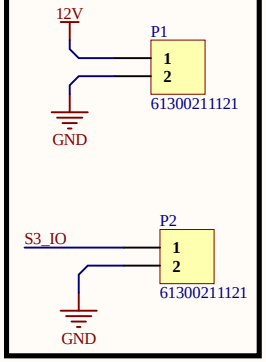
UART SIGNALS



PROGRAMMING HEADER



S3 SIGNALS

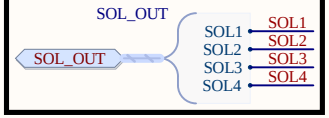


P1 has the 12V input and output supply signals from S3 board.

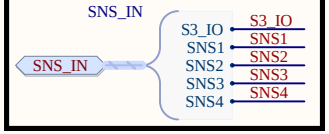
S3_IO is the 12V interrupt signal from S3 board.



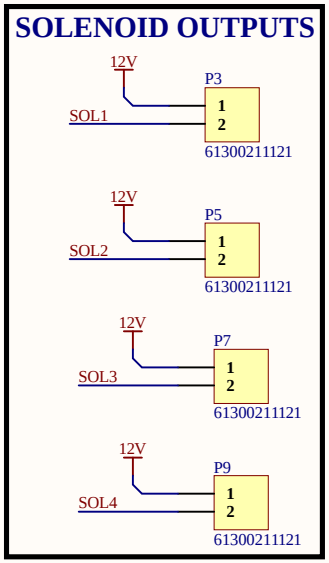
SOLENOID OUTPUT



LEVEL SENSE INPUT

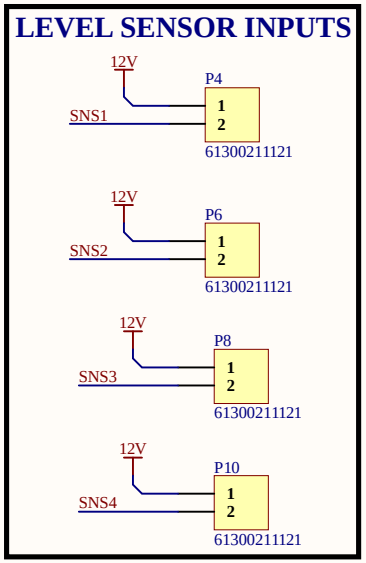


SOLENOID OUTPUTS

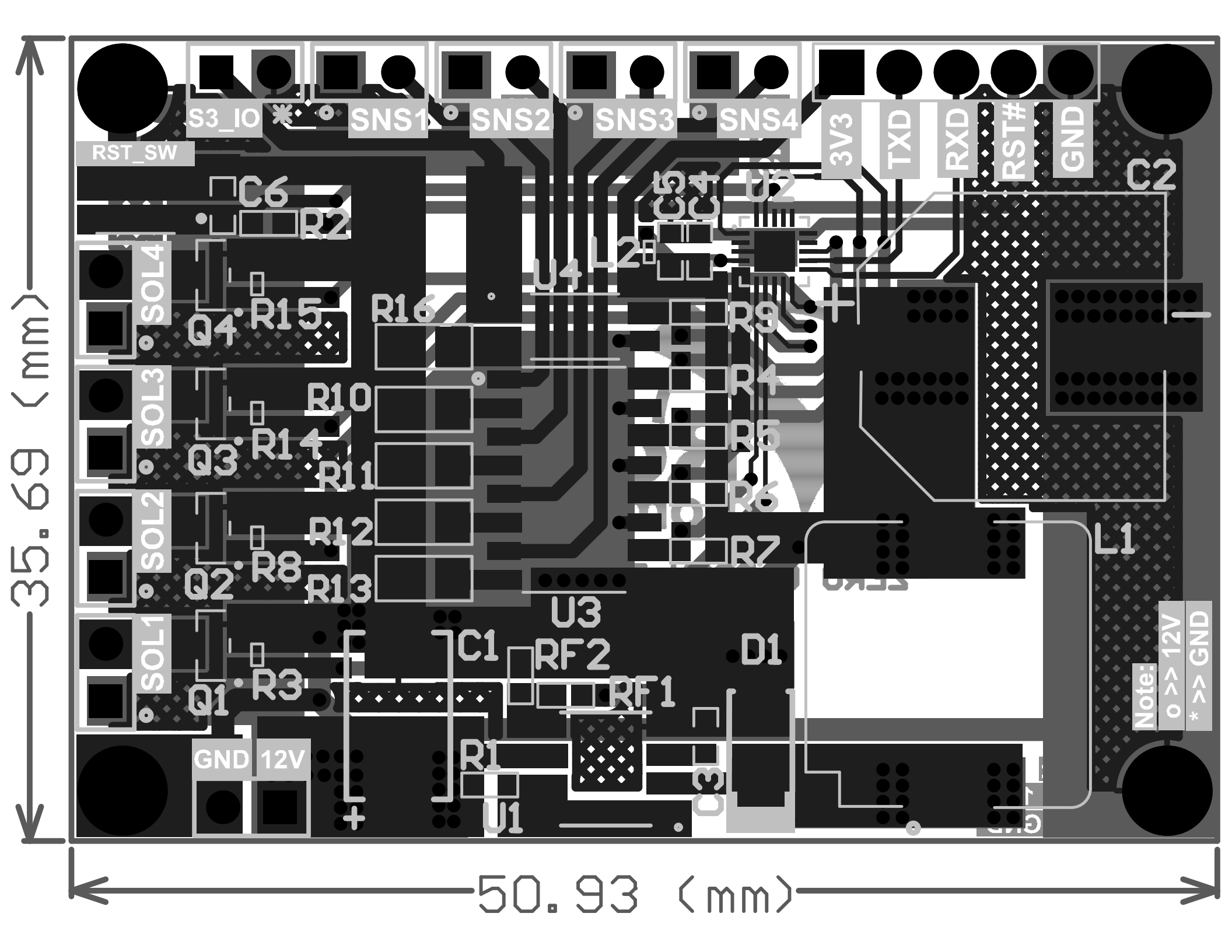


Solenoid GND controlled by MCU via MOSFETS. When any of the level sensors is closed, the respective solenoid is excited in response.

LEVEL SENSOR INPUTS



Level Sensor will act as a switch and operate on 12V. When the water level is sensed, the switch closes indicating the change. This change will be sensed by optocouplers.


S3_IO
SNS1
SNS2
SNS3
SNS4
3V3
TXD
RXD
RST#
GND
RST_SW
C6
R2
C4
C5
U2
L2
U4
C2
SOL4
Q4
R15
R16
R9
R4
R5
R6
R7
SOL3
Q3
R14
R10
R11
SOL2
Q2
R8
R12
R13
U3
L1
SOL1
Q1
R3
C1
RF2
RF1
D1
R1
GND
12V
U1
Note:
o -> 12V
* -> GND
35.69 (mm)
50.93 (mm)

Comment	Description	Designator	Footprint	LibRef	Quantity
100uF	CAP TANT 100UF 20% 25V 2917	C1	CAP- 2917	TAJE107M025RNJ	1
470uF	470uF ±20% 50V 12.5mm 13.5mm 1000hrs 105°C -40°C~+105°C SMD,12.5x13.5mm Aluminum Electrolytic Capacitors - SMD RoHS	C2	VT1H471MI135000CE0	VT1H471MI135000CE0	1
470pF	CAP CER 470pF 10% X7R 50V 0603 SMD	C3	Cap0603	CC0603KRX7R9BB471	1
0.1uF	CAP CER 0.1UF 50V Y5V 0603	C4, C6	Cap0603	CC0603ZRY5V9BB104	2
1uF	CAP CER 1UF 25V X5R 0603	C5	Cap0603	CC0603KRX5R8BB105	1
SS24A	40V 500mV@2A 2A SMA(DO-214AC) Schottky Barrier Diodes (SBD) ROHS	D1	SMA	SS24A	1
PZ254V-11-05P	Pin Header 5 1 right-angle	J1	HDR1X5	PZ254V-11-05P	1
220uH	Inductor Power Shielded Wirewound 220uH 20% 1KHz Ferrite 1.3A 310mOhm DCR Blister T/R	L1	SLF12575T-221M1R3-PF	SLF12575T-221M1R3-PF	1
MMZ0603S601CT000	FERRITE BEAD 600 OHM 0201 1LN	L2	IND 0201	MMZ0603S601CT000	1
61300211121	CONN HEADER VERT 2POS 2.54MM	P1, P2, P3, P4, P5, P6, P7, P8, P9, P10	HDR1X2	Header 1x2	10
DMN13H750S-7	MOSFET N-CH 130V 1A SOT23	Q1, Q2, Q3, Q4	SOT23-3	DMN13H750S-7	4
0.33R	0.33Ω 0603 Chip Resistor - Surface Mount ROHS	R1	RES-0603	RTT03R330FTP	1
4.7K	RES SMD 4.7K OHM 1% 1/10W 0603	R2	RES-0603	RC0603FR-074K7L	1
10K	RES SMD 10K OHM 1% 1/20W 0201	R3, R8, R14, R15	Res-0201	RC0201FR-0710KL	4
3.3K	RES SMD 3.3K OHM 1% 1/10W 0603	R4, R5, R6, R7, R9	RES-0603	RC0603FR-073K3L	5
536R	Chip Resistor Automotive Grade Thick Film 1206 536Ohm 1% Paper T/R, RES SMD 1206 536Ohm 1% Paper T/R	R10, R11, R12, R13, R16	Res-1206	AC1206FR-07536RL	5
2K	RES SMD 2K OHM 1% 1/10W 0603	RF1	RES-0603	RC0603FR-072KL	1
1.2K	±1% 1/10W Thick Film Resistors 75V ±100ppm/°C -55°C~+155°C 1.2kΩ 0603 Chip Resistor - Surface Mount ROHS	RF2	RES-0603	0603WAF1201T5E	1
TS-1185EC-C-D-B	3mm 2mm Round Button 50mA Top3.65mm SPST 12V SMD Tactile Switches	SW1	TS-1185EC-C-D-B	TS-1185EC-C-D-B	1
MC34063ADR	1.5-A boost, buck, inverting switching regulator, operation from 0°C to 70°C 8-SOIC 0 to 70	U1	SOIC-8	MC34063ADR	1
ATTINY1616-MNR	IC MCU 8BIT 16KB FLASH 20VQFN	U2	20-QFN	ATTINY1616-MNR	1
LTV-247	OPTOISO 3.75KV 4CH TRANS 16SO	U3	LTV-247	LTV-247	1
PS2801C-1-F3-A	TransistorOptocouplers DC 1 2500Vrms SOP-4_P1.27 Optocouplers RoHS	U4	PS2801C-1-F3-A	PS2801C-1-F3-A	1